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Decoding the Job Market

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1 Abstract

This paper investigates whether the relationships between key labor market indicators and unemployment changed following the COVID-19 pandemic. Using monthly U.S. data from the Bureau of Labor Statistics, we analyze the three-month change in the unemployment rate as a function of indicators capturing vacancies, worker flows, separations, labor force attachment, and underemployment. The sample is divided into pre-COVID (before 2020), COVID (2020–2021), and post-COVID (2022 onward) regimes, with primary focus on differences between the pre- and post-pandemic periods.

We combine correlation analysis, Beveridge Curve comparisons, and multivariate OLS regressions estimated with Newey–West heteroskedasticity and autocorrelation consistent standard errors. Results indicate a post-2021 reconfiguration of labor market co-movements and a shift in the vacancy–unemployment relationship. In regression analysis, measures of involuntary labor market slack remain robustly associated with unemployment changes in the post-COVID period. The findings suggest that the structure of unemployment dynamics evolved after the pandemic, with implications for how labor market conditions are interpreted.

2 Introduction

As college students preparing to enter the workforce, we have a practical question in mind: when the news says the job market is "strong" or "weak," what signals should we actually watch? Many students and early-career applicants describe a market that feels more selective and more competitive—longer recruiting timelines, more interview rounds, and more applicants per opening—even during periods when headline unemployment remains relatively low. Rather than relying on personal anecdotes or social media narratives, this project uses standard labor market indicators to evaluate whether measurable patterns in the data support these perceptions and whether the relationships behind unemployment have shifted across recent economic periods.

Unemployment is the most visible labor market statistic, but it is only one dimension of labor market conditions. Employers and job seekers respond to job openings, worker turnover, layoffs, and labor force participation in ways that unemployment alone cannot capture. In this study, we examine unemployment in conjunction with the job openings rate, quits rate, layoffs and discharges rate, initial unemployment claims, labor force participation rate, part-time employment for economic reasons, permanent job losers, and wage growth. Together, these indicators capture both demand-side behavior (vacancies and separations) and worker-side outcomes (job loss, underemployment, and attachment to the labor force). Definitions and measurement follow the Bureau of Labor Statistics (BLS) Current Population Survey and Job Openings and Labor Turnover Survey documentation (U.S. Bureau of Labor Statistics, 2025, 2013).

To connect these indicators to an intuitive picture of the job market, we use the Beveridge Curve framework, which describes the relationship between unemployment and vacancies. Movements along the curve are typically associated with cyclical changes in labor demand, while shifts in the curve can indicate changes in matching efficiency or broader structural conditions (Blanchard and Diamond, 1989; Petrongolo and Pissarides, 2001). Because the COVID-19 pandemic produced a sharp disruption in 2020 and the recovery period unfolded unevenly through 2021 and beyond, recent work has documented notable changes in Beveridge-curve dynamics after the pandemic (Figura and Waller, 2024; Kassens and Rodgers, 2022). In practical terms, this framework helps translate abstract macroeconomic concepts into something students care about: whether vacancies translate into hires as efficiently as before, and whether low unemployment still implies an "easy" job search.

Our empirical strategy emphasizes the comparison most relevant to current job seekers: the pre-2020 (pre-COVID) period versus the post-2021 (post-COVID) period, while treating 2020–2021 as a distinct COVID regime to separate shock-driven behavior from recovery-period patterns. We focus on changes rather than levels by modeling unemployment as a three-month change in the unemployment rate (in percentage points), and we transform explanatory indicators using change measures appropriate to each series. For variables already expressed as rates, we use three-month percentage-point changes; for level-like series, we use three-month percent changes. Because job openings, quits, and initial claims are often used as leading measures when people informally monitor the labor market, we incorporate a one-month lead for these indicators.

Finally, to ensure the results are statistically reliable for monthly time series, we estimate multivariate OLS regressions with Newey–West HAC standard errors (including small-sample correction) (Newey and West, 1987; Andrews, 1991). We then use pooled interaction models and Wald tests to evaluate whether the strength of each indicator's relationship with unemployment differs across periods. The goal is not to build a forecasting tool, but to identify which indicators provide the clearest and most consistent signals about labor market stress—information that can help students interpret job market headlines and understand what is changing beneath the surface.

3 Empirical Framework

3.1 Data and Transformations

We utilize monthly U.S. labor market time series from the Federal Reserve Economic Data (FRED) database, sourced from the Bureau of Labor Statistics (BLS). Our dependent variable, Y_t, represents the three-month percentage-point change in the unemployment rate (UNRATE):

$$Y_t = \text{UNRATE}_t - \text{UNRATE}_{t-3} \tag{1}$$

To maintain consistency across measurement units, we apply the transformations outlined in Table 1. Level-like series are converted to three-month percent changes, while existing rates are measured as three-month percentage-point changes. Additionally, we lead job openings, quits, and initial claims by one month to align with real-time monitoring practices.

Table 1: Explanatory Indicators and Transformations

Variable	**Code**	**Transformation**
Job Openings Rate	JTSJOR	3-mo Δ (pts), Lead 1-mo
Quits Rate	JTSQUR	3-mo Δ (pts), Lead 1-mo
Layoffs and Discharges	JTSLDR	3-mo Δ (pts)
Initial Claims	ICSA	3-mo % Δ, Lead 1-mo
Labor Force Participation	CIVPART	3-mo Δ (pts)
Part-time for Econ. Reasons	`PART_TIME_ECON`	3-mo % Δ
Permanent Job Losers	`PERM_JOB_LOSER`	3-mo % Δ
Wage Growth	–	3-mo Δ (pts)

3.1.1 Analysis Regimes

To contextualize shock-driven dynamics, we analyze the following three regimes:

- **Pre-COVID:** Observations prior to 2020-01-01.
- **COVID:** 2020-01-01 through 2021-12-31.
- **Post-COVID:** Observations on or after 2022-01-01.

Our primary comparative analysis focuses on the Pre-2020 versus Post-2021 periods.

3.2 Methods

We apply three complementary methods. First, we compute Pearson correlation matrices of the transformed indicators for each regime to assess whether co-movement patterns shift across periods.

Second, we visualize the Beveridge Curve relationship between unemployment and job openings and examine whether the post-COVID period displays a visible shift relative to the pre-COVID relationship. The Beveridge Curve provides a standard framework for interpreting vacancy–unemployment dynamics and potential changes in matching efficiency (Blanchard and Diamond, 1989; Petrongolo and Pissarides, 2001), and recent work has discussed how these dynamics shifted following the pandemic (Figura and Waller, 2024; Kassens and Rodgers, 2022).

Third, we estimate multivariate OLS regressions within each regime:

$$Y_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \cdots + \beta_k X_{k,t} + \varepsilon_t,$$

where Y_t is the three-month change in unemployment and $X_{i,t}$ are the transformed indicator changes (including the specified one-month lead for selected indicators). Because monthly labor market changes can exhibit autocorrelation and heteroskedasticity, we report Newey–West HAC standard errors with small-sample correction (Newey and West, 1987; Andrews, 1991). To test whether relationships differ across regimes, we fit a pooled interaction model and conduct Wald tests that evaluate whether coefficients are statistically different between pre-COVID and post-COVID periods.

3.3 Results

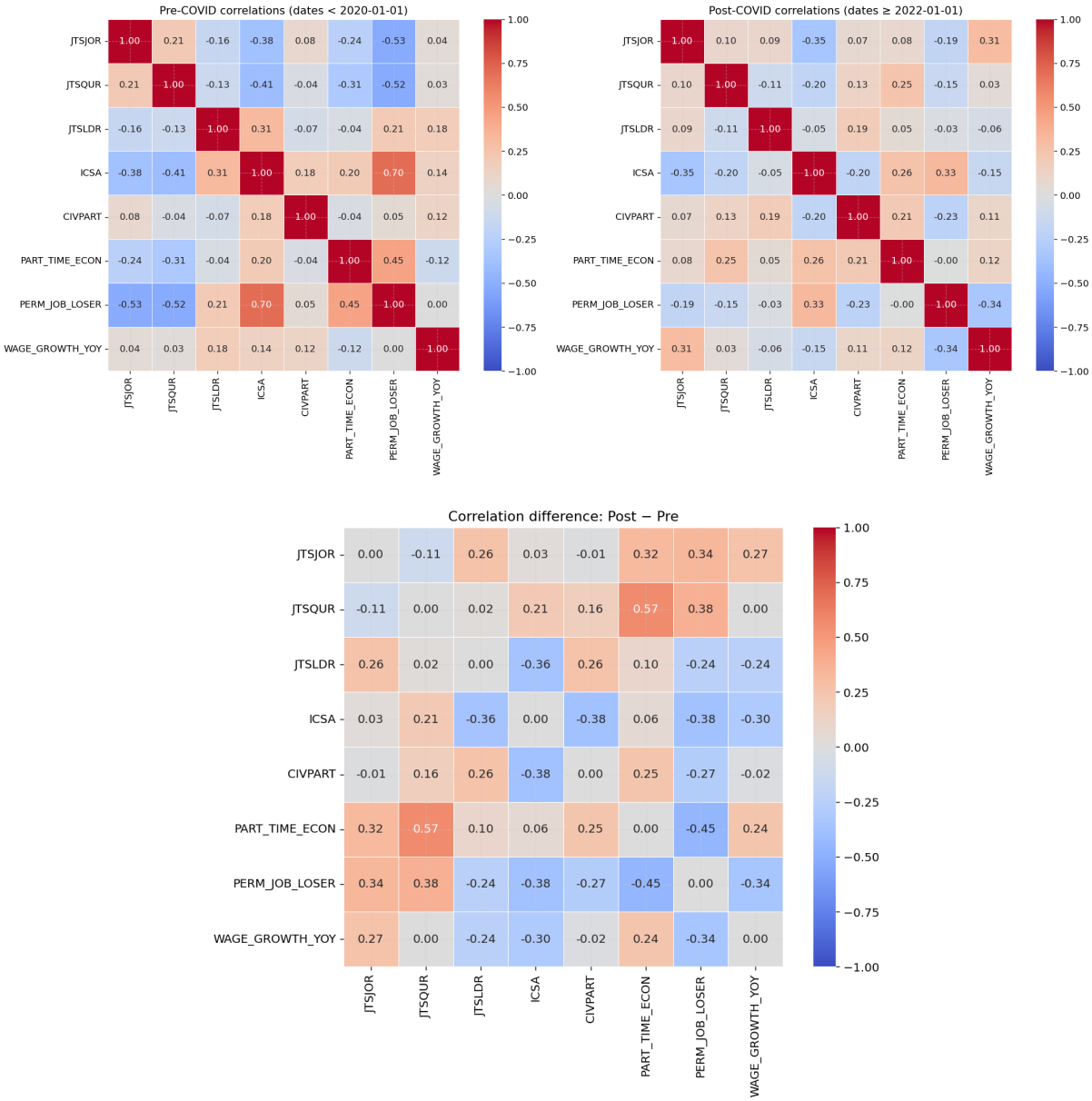

Figure 1: Correlation structure across regimes. Top left: Pre-COVID. Top right: Post-COVID. Bottom: Change in correlations (post minus pre).

Correlation matrices show that co-movement among labor market indicators differs across regimes, with the post-COVID period exhibiting weaker and differently patterned relationships relative to the pre-COVID period.

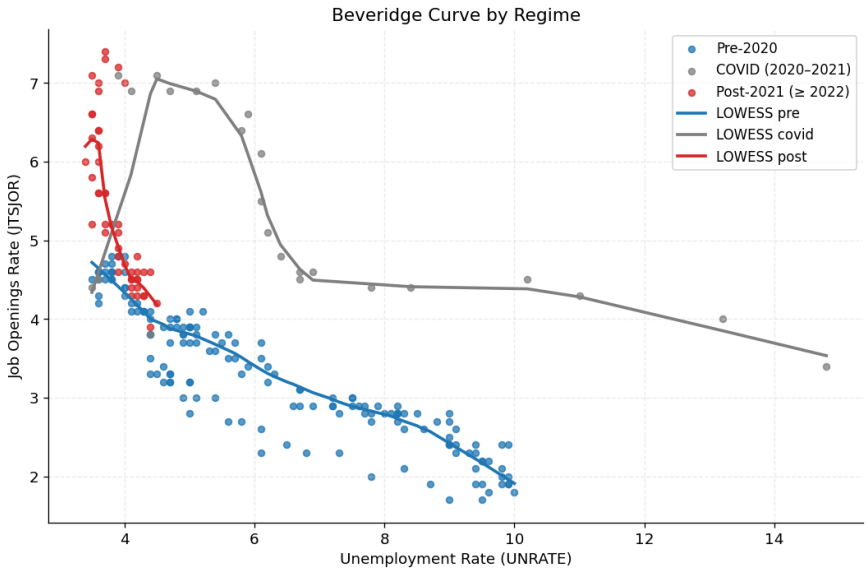

Figure 2: Beveridge curve projections before, after, and during COVID.

The Beveridge Curve plot indicates a visible post-COVID shift in the vacancy–unemployment relationship, consistent with recent research documenting changes in matching dynamics (Figura and Waller, 2024; Kassens and Rodgers, 2022).

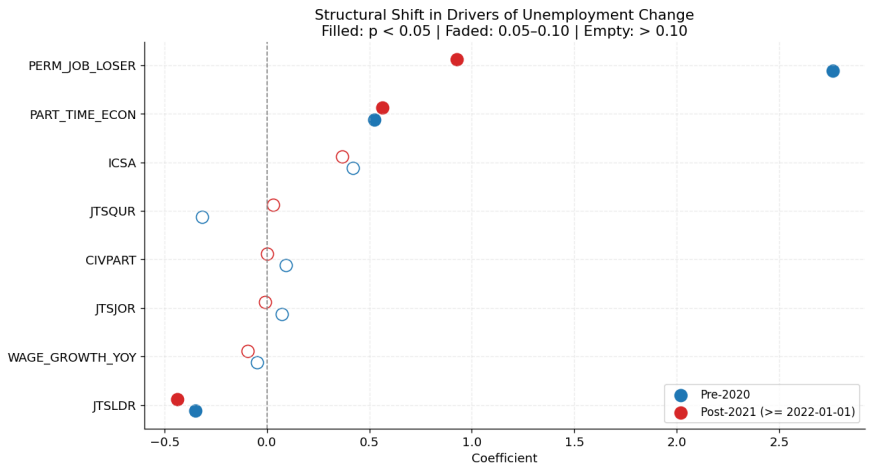

Figure 3: Coefficients before and after COVID

Multivariate OLS results with HAC inference suggest that job-loss and underemployment measures remain the most robust associates of unemployment changes in the post-COVID period, while pooled interaction regressions and Wald tests are used to formally assess whether these coefficients differ statistically across regimes.

4 Conclusion

This project began with a practical question that many college students and early-career applicants face: when people say the job market is "strong" or "weak," what signals actually describe the market that job seekers experience? Our results suggest that the most useful insight comes from identifying which forces are driving unemployment changes and whether those forces behave differently in the post-2021 economy compared to the pre-2020 period.

The Beveridge Curve figure helps explain why perceptions of a difficult job search can persist even when unemployment is relatively low. In the post-2021 period, job openings remain elevated at comparable unemployment rates, and the smoothed post-2021 relationship sits above the pre-2020 curve. This pattern is consistent with an outward shift in the vacancy–unemployment relationship, which is often interpreted as evidence that matching workers to jobs has become less straightforward—for example due to skill mismatch, sectoral reallocation, geographic constraints, or more selective and slower hiring processes. For students, the implication is simple: high vacancies do not necessarily mean hiring will feel easy.

The coefficient comparison figure reinforces which indicators most directly track unemployment changes in each regime. In both the pre-2020 and post-2021 periods, the most robust associations are tied to worker-side slack: permanent job losers and part-time employment for economic reasons are positively related to changes in unemployment. In contrast, vacancy-side variables such as job openings and quits cluster near zero in the coefficient plot and do not appear to add reliable incremental explanatory power in the multivariate specification once job-loss and underemployment measures are included.

Finally, the most important structural insight in the pre vs. post comparison is that permanent job losers behave differently after 2021. This supports a narrative in which post-2021 unemployment increases may be more closely linked to separations that are harder to reverse—jobs that disappear or restructure rather than temporarily pause—and therefore translate more directly into sustained unemployment changes when matching frictions are elevated. Taken together, these results imply that the post-2021 labor market is not simply "better" or "worse" than the pre-2020 market, but more structurally complex. For students trying to interpret labor market conditions, monitoring indicators of involuntary slack (especially permanent job loss and involuntary part-time work) alongside the Beveridge Curve provides a clearer, more grounded read of job-market stress than unemployment or vacancy headlines alone.

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